) Show that $a+a=a \ \forall a \in B$

By axiam 4 $a = a+0$

$= a+a \cdot a'$

$= (a+a) \cdot (a+a')$

$= (a+a) \cdot 1$

$a+a$

$a+a = a \ \forall a \in B$

) Show that $a \cdot a = a \ \forall a \in B$

By axiam 4 $a = a \cdot 1$

$= a \cdot (a+a')$

$= (a \cdot a) + (a \cdot a')$

$= a \cdot a + 0$

$= a \cdot a$

Thus $a \cdot a = a \ \forall a \in B$

Show that elements 0 & 1 boolean of algebra are unquie

Assume $0_1, 0_2$ be two zero elements

$0_2 + 0_1 = 0_2$ taking 0_1 as zero elment

$0_1 + 0_2 = 0_1$ taking 0_2 as zero elment

By commutative law

$0_1 + 0_2 = 0_2 + 0_1$

$0_1 = 0_2$

$\therefore$ zero is unique

ii) Let $1_1, 1_2$ be two unit elements

$1_1 \cdot 1_2 = 1_1$ taking 1_2 as unit element

$1_2 \cdot 1_1 = 1_2$ taking 1_1 as unit element

By commutativity,

$$1_1 \cdot 1_2 = 1_2 \cdot 1_1$$

$$1_1 = 1_2$$

$\therefore$ Identity element is unique

4) If B is a boolean algebra, then for all $a \in B$ demonstrate that $a+1=1$

Given B is boolean algebra Let $a \in B$

$$\begin{aligned} a+1 &= (a+1) \cdot 1 \\ &= 1 \cdot (a+1) \\ &= (a+a')(a+1) \\ &= a + (a' \cdot 1) \\ &= a + a' = 1 \end{aligned}$$

$\therefore a+1 = 1 \ \forall a \in B$

5) If B is a Boolean algebra then prove $a \cdot 0 = 0$

$$\begin{aligned} a \cdot 0 &= a \cdot 1' \\ &= a \cdot (a+1)' \\ &= a \cdot (a' \cdot 1') \\ &= (a \cdot a') \cdot 1' \\ &= (a \cdot a') \cdot 0 \\ a \cdot 0 &= 0 \ \forall a \in B \end{aligned}$$

$= 0 \cdot 0 = 0$

Date

6) Show that $(a+b)\cdot(a'+c) = ac + a'b + bc$

$$(a+b)\cdot(a'+c) = (a+b)\cdot a' + (a+b)\cdot c$$
$$= a\cdot a' + b\cdot a' + a\cdot c + b\cdot c$$
$$= 0 + a'\cdot b + a\cdot c + b\cdot c$$
$$= ac + a'b + bc$$

7) <u>Boundedness law</u>

Also known as dominant law. It states
$a+1 = 1$ and $a\cdot 0 = 0$ $\forall\, a \in B$

8) In a boolean algebra, for any $a, b \in B$ show
$a+b' = 1$ if $a+b = a$

assume $a+b' = 1$
$a+b = a$
$$a+b = (a+b)\cdot 1$$
$$= (a+b)(a+b')$$
$$= a+b\cdot b' = a+0 = a$$

also $a+b = a$
$a+b' = 1$
$$a+b' = 1\cdot(a+b')$$
$$= (a+a')\cdot(a+b')$$
$$= a + (a'\cdot b')$$
$$= a + (a+b)'$$
$$= a + a'$$
$$= 1$$

$\therefore$ Hence proved

9) Boolean algebra

A non empty set B with two binary operation + and · on B, a unary operation & two distinct elements 0 and 1 is called Boolean algebra denoted by $(B, +, \cdot, ', 0, 1)$ and it should satisfy

→ commutative → distributive → complement
→ Associative → Identity

10) Commutative law

$a + b = b + a$ and $a \cdot b = b \cdot a$

11) State associative law

$a + (b + c) = (a + b) + c$
$a \cdot (b \cdot c) = (a \cdot b) \cdot c$

12) distributive law

$a + (b \cdot c) = (a + b) \cdot (a + c)$
$a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

13) Identity law

There exists $0, 1 \in B$ such that
$a + 0 = a$ and $a \cdot 1 = a$

14) complement law

For each $a \in B$ there exist an element $a' \in B$ such that

$a + a' = 1$ and $a \cdot a' = 0$

iv) In boolean algebra B, $0' = 1$ & $1' = 0$

Soln $0' = 0 + 0'$

$= 1$

$1' = 1' \cdot 1$

$= 0$

$\therefore$ Hence proved

10 m

1) In a boolean (i) $a+b = b$ (iii) $a' + b = 1$

(ii) $a \cdot b = a$ (iv) $a \cdot b' = 0$

T.P $a+b = b$

assume $a \cdot b = a \cdot (a+b)$

$a \cdot b = a$

$\therefore$ Hence proved

T.P $a \cdot b = a$

assume $a' + b = 1$

$a' + a + b' = a' + b$

$a' + 1 = a' + b$

$a' + 0' =$

$(a \cdot 0)'$

$0' = 1$

$1 = a' + b$

$\therefore$ Hence proved

T.P $a' + b = 1$

assume $a \cdot b' = 0$

$(a' + b)' = 1'$

$(a')' \cdot b' = 1'$

$a \cdot b' = 0$

$\therefore$ Hence proved

T.P $a \cdot b' = 0$

assume $a + b = b$

$(a \cdot b')' = 0$

$a' + (b')' = 0'$

$a' + b = 1$

$a + b = (a+b) \cdot 1$

$= (a+b) \cdot (a'+b)$

$= (b + a)(b + a')$

$= b + a \cdot a'$

$= b + 0 = b$

$\therefore$ Hence proved

i) De morgans law $(a+b)' = a' \cdot b' \ \forall a, b \in B$

Let x' be the complement of x then

$$x + x' = 1 \text{ and } x \cdot x' = 0$$

$$(a+b) + a' \cdot b' = 1$$

$$(a+b) \cdot (a' \cdot b') = 0$$

consider $(a+b) + a' \cdot b' = [(a+b)+a'] \cdot [(a+b)+b']$

$$= [(b+a)+a'] \cdot [a+(b+b')]$$

$$= [b+(a+a')] \cdot [a+1]$$

$$= (b+1) \cdot (a+1)$$

$$= 1 \cdot 1 = 1$$

$(a+b) \cdot (a' \cdot b') = (a' \cdot b')(a+b)$

$$= (a' \cdot b') \cdot a + (a' \cdot b') \cdot b$$

$$= a \cdot (a' \cdot b') + a' \cdot (b' \cdot b)$$

$$= (a \cdot a') \cdot b' + a' \cdot (b \cdot b')$$

$$= 0 \cdot b' + a' \cdot 0$$

$$= b' \cdot 0 + a' \cdot 0$$

$$= 0 + 0$$

$$= 0$$

Thus $(a+b) + a' \cdot b' = 1$

$$(a+b) \cdot (a' \cdot b') = 0$$

Hence $a' \cdot b'$ is complement of $a+b$

$$(a+b)' = a' \cdot b'$$

In a boolean algebra $ab' + a'b = 0$ if & only if $a = b$.

assume $a = b$

T.P $ab' + a'b = 0$

$ab' + a'b = bb' + a'a$

$= 0 + 0$

$= 0$

let $ab' + a'b = 0$

$a = b$

Here $a = a \Rightarrow a + 0 = a$

$a + (ab' + a'b) = a$

$(a + ab') + a'b = a$

$a + a'b = a$

$(a + a') \cdot (a + b) = a$

$1 \cdot (a + b) = a$

$a + b = a \longrightarrow (1)$

similarly $b = b \Rightarrow 0 + b = b$

$\Rightarrow ab' + a'b + b = b$

$\Rightarrow ab' + b = b$

$\Rightarrow (a + b) \cdot (b' + b) = b$

$\Rightarrow (a + b) \cdot 1 = b$

$\Rightarrow a + b = b \longrightarrow (2)$

From (1) and (2) we get $a = b$

4) P.T $a \cdot b' = 0$ if $a \cdot b = a$

Assume $a \cdot b' = 0$

T.P $a \cdot b = a$

$a \cdot b' = 0$

$(a \cdot b')' = 0'$

$a' \cdot (b')' = 1$

$a' + b = 1$

$a \cdot (a' + b) = a \cdot 1$

$(a \cdot a') + (a \cdot b) = a$

$0 + a \cdot b = a$

$a \cdot b = a$

$a \cdot b = a$

$a' + b' = a'$

$(a \cdot b)' = a'$

$a \cdot (a' + b') = a \cdot a'$

$(a \cdot a') + (a \cdot b') = a \cdot a'$

$0 + a \cdot b' = 0$

$a \cdot b' = 0$

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5) state & prove demorgan law

(i) $(a + b)' = a' \cdot b'$

Let x' be complement of x

then $x + x' = 1$ & $x \cdot x' = 0$

$(a + b) + a' \cdot b' = 1$

and $(a + b) \cdot (a' \cdot b) = 0$

consider $(a + b) + a' \cdot b' = [(a+b) + a'] \cdot [(a+b) + b']$

$= [(b+a) + a'] \cdot [a + (b + b')]$

$= [b + (a + a')] \cdot [a + 1]$

$= (b + 1) \cdot (a + 1)$

$= 1 \cdot 1 = 1$

$(a + b) \cdot (a' \cdot b') = (a' \cdot b')(a + b)$

$= (a' \cdot b') \cdot a + (a' \cdot b') \cdot b$

$= a \cdot (a' \cdot b') + a' \cdot (b' \cdot b)$

$= (a \cdot a') \cdot b' + a' \cdot (b \cdot b')$

$= 0 \cdot b' + a' \cdot 0$

$= b' \cdot 0 + a' \cdot 0$

$= 0 + 0 = 0$

Thus $(a+b) + a' \cdot b' = 1$

$(a+b) \cdot (a' \cdot b') = 0$

$(a+b)' = a' \cdot b'$

(ii) $a \cdot b$ is $a' + b'$ we have to prove

$(a \cdot b) + (a' + b') = 1$

$(a \cdot b)(a' + b') = 0$

$(a \cdot b) + (a' + b') = [a + (a' + b')][b + (a' + b')]$

$= [(a + a') + b'][b + (b' + a')]$

$= [(a + a') + b'] \cdot [(b + b') + a']$

$= [1 + b'] \cdot [1 + a']$

$= [b' + 1] \cdot [a' + 1]$

$= 1 \cdot 1 = 1$

$(a \cdot b) \cdot (a' \cdot b') = (a \cdot b) \cdot a' + (a \cdot b) \cdot b'$

$= (b \cdot a) \cdot a' + a \cdot (b \cdot b')$

$= b \cdot (a \cdot a') + a \cdot (b \cdot b')$

$= b \cdot 0 + a \cdot 0$

$= 0 + 0 = 0$

6) <u>Boundedness law</u>

(i) $a + 1 = 1$ (ii) $a \cdot 0 = 0$

$a + 1 = a + (a + a')$

$= (a + a) + a'$

$= a + a'$

$= 1$

$= a \cdot 0 + 0$

$= a \cdot (0 + a')$

$= a \cdot a'$

$= 0$

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7) <u>Idempotent law</u>

$a + a = a$ $a \cdot a = a \quad \forall a \in B$

$a = a + 0$

$= a + a \cdot a'$

$= (a + a) \cdot (a + a')$

$= (a + a) \cdot 1$

$= a + a$

Thus $a + a = a \; \forall a \in B$

$a = a \cdot 1$

$= a \cdot (a + a')$

$= a \cdot a + a \cdot a'$

$= a \cdot a + 0$

$= a \cdot a$

$a \cdot a = a \; \forall a \in B$

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8) In a boolean algebra B, complement of every element is unique

a_1' & a_2' be two complement of a

$a + a_1' = 1$ & $a \cdot a_1' = 0$

$a + a_2' = 1$ & $a \cdot a_2' = 0$

$a_2' = a_2' \cdot 1$

$= a_2' \cdot (a + a_1')$

$= a_2' \cdot a + a_2' \cdot a_1'$

$= a \cdot a_2' + a_1' \cdot a_2'$

$= 0 + a_1' \cdot a_2'$

$= a \cdot a_1' + a_1' \cdot a_2'$

$= a_1' \cdot a + a_1' \cdot a_2'$

$= a_1' \cdot (a + a_2')$

$= a_1' \cdot 1$

$a_2' = a_1'$

$\therefore$ complement of element is unique